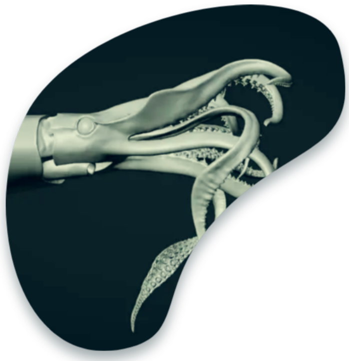

Assessment of Jumbo Squid in the SE Pacific — State-Space Biomass Dynamics Model & ENSO Impacts

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Category: Working paper summary

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1 Document

A stock assessment paper by China (Li and Chen 2025) (Gang Li & Xinjun Chen, National Data Centre for Distant-water Fisheries of China / Shanghai Ocean University), presented as SC13-SQ07 to the 13th Meeting of the SPRFMO Scientific Committee (8–13 September 2025, Wellington, New Zealand).

2 Subject

A regional stock assessment of jumbo squid (*Dosidicus gigas*) in the Southeast Pacific, combining catch and multi-fleet abundance indices with environmental drivers. Its distinguishing feature is an explicit test of how the El Niño–La Niña (ENSO) cycle affects population dynamics, treating *D. gigas* as a single stock unit across the region (per recent genetic evidence).

3 Methods

A **Bayesian state-space surplus production (Pella-Tomlinson) model**, fitted in WinBUGS via MCMC. Four abundance indices were used: Peruvian and Chilean CPUE from CALAMASUR, plus the Chinese squid-jigging fleet CPUE standardized with a **Generalized Additive Model (GAM)** using a gamma distribution and environmental covariates (SST, chlorophyll-a, salinity, Niño 1+2 index) and spatiotemporal terms. Catch data (2012–2024) came from FAO plus national statistics. **Seven environment-dependent models** were tested, letting carrying capacity K, intrinsic growth r, and shape s vary between El Niño and La Niña phases (defined by Niño 1+2 SST anomalies). One base scenario plus three prior-sensitivity scenarios were run.

4 Key results

- The GAM confirmed year, month and salinity as the main CPUE drivers; nominal and standardized CPUE both peaked in 2015 and were lowest in 2024.
- Among the seven environment-dependent models, the **Kr model** (both K and r vary with ENSO) had the lowest DIC and was best supported — climate drives fluctuations in both carrying capacity and growth rate.
- Base-scenario MSY **1.19–1.35 million tonnes** (traditional model); the environment-dependent model gave MSY around 1.3–1.6 Mt, differing between El Niño and La Niña phases.
- B_{2012}/K 0.5, indicating the stock was at roughly half of unexploited biomass at the start of the series.

- **Kobe-plot diagnostics show the stock has been sustainable since 2013** — no overfishing and not overfished. Terminal-year probability of healthy status was 58.6–73.1% (traditional model) and **70.9–88.3% (environment-dependent model)**.

5 Main conclusion

The stock is **neither overfished nor undergoing overfishing**, and has remained healthy since 2013 with catch consistently below MSY. Notably, despite the sharp decline in *D. gigas* landings in 2024, the stock stayed healthy — a pattern the authors link to the influence of El Niño on population dynamics. The environment-dependent (Kr) model reinforces that climate-driven variability in carrying capacity and growth rate is central to the stock's behaviour.

6 Way forward

Continue integrating environmental (ENSO) drivers into the assessment, given the strong sensitivity of carrying capacity and growth rate to climate. Further refinement of the standardized CPUE indices and the environment-dependent model structure would improve reference points and terminal-year status estimates.

References

Li, Gang, and Xinjun Chen. 2025. *Assessment of Jumbo Squid in Southeast Pacific Ocean Based on State-Space Biomass Dynamics Model and Impacts of ENSO.* Scientific Committee Document Nos. SC13-SQ07. National Data Centre for Distant-water Fisheries of China, Shanghai Ocean University.